Group ID	3
Student ID	**230042106, 230042117, 230042143**
Title	1757
Brief Description	1757 is a real-time, web-based multiplayer social deduction game for 6 to 15 players inspired by the historical events of the Battle of Plassey (1757). Built on a Node.js, Socket.io, and Redis backend with a React Native frontend, the application implements a strict finite state machine for the game loop (Discussion, Selection, Ballot, and Resolution) featuring dynamic role-based UI interfaces, real-time voting, private encrypted communication channels for spies, and advanced character roles.

Serial	FR ID	Functional Requirement	User Story ID	US Title	As a (precise role)	I want (requirement)	So that (completion condition)	User Story	Acceptance Criteria	Priority	Status
1	FR_1	The system shall support real-time lobby creation and customization.	US_1	Create Private Lobby	Lobby Host	to create a private game lobby with a unique room code	I can invite my friends to join a secure game session	As a Lobby Host, I want to create a private game lobby with a unique room code, so that I can invite my friends to join a secure game session.	- A unique 6-character alphanumeric room code is generated. - The host can set the maximum player limit between 6 and 15 players. - A "Copy Invite Link" button copies the room code to the clipboard. - The lobby displays the real-time count of connected players.	High	Done
2	FR_1	The system shall support real-time lobby creation and customization.	US_2	Lobby Game Settings	Lobby Host	to configure the active roles and deck setup in the lobby	I can customize the gameplay difficulty and rules before starting the match	As a Lobby Host, I want to configure the active roles and deck setup in the lobby, so that I can customize the gameplay difficulty and rules before starting the match.	- Toggle switches are available for advanced roles (e.g., Jagath Seth, de Bussy). - The host can adjust the discussion timer duration (30s to 300s). - The settings dynamically synchronize to all connected players in the lobby in real time.	Medium	To Do
3	FR_2	The system shall manage player matchmaking and room joining.	US_3	Public Matchmaking Join	Player	to join a public matchmaking room via a global room list	I can play the game with other active online players without needing an invite link	As a Player, I want to join a public matchmaking room via a global room list, so that I can play the game with other active online players without needing an invite link.	- A scrollable list of open public lobbies is displayed with active player counts. - A "Quick Join" button automatically places the player into the next available lobby. - The matchmaking screen auto-refreshes every 60 seconds.	High	In Progress
4	FR_3	The system shall support real-time public communication during gameplay.	US_4	Public Court Chat	Player	to send text messages in the public court chat channel	I can debate, share suspicions, and coordinate with other players	As a Player, I want to send text messages in the public court chat channel, so that I can debate, share suspicions, and coordinate with other players.	- A chat input field is available at the bottom of the screen. - Sent messages appear immediately on all players' screens with their avatar and username. - Messages are filtered for offensive words using a client-side library.	Medium	In Progress
5	FR_4	The system shall distribute secret roles and adapt the interface accordingly.	US_5	Reveal Secret Role	Player	to view my secret role and faction alignment at the start of the match	I know my objective and can plan my strategy for the game	As a Player, I want to view my secret role and faction alignment at the start of the match, so that I know my objective and can plan my strategy for the game.	- A card flip animation reveals the role (e.g., Loyalist or EIC Spy) upon clicking a "Reveal" card. - A role description helper explains the role's victory conditions.	High	To Do
6	FR_5	The system shall reveal faction-specific metadata to authorized roles.	US_6	Siraj ud-Daulah Saboteur Vision	Siraj ud-Daulah	to see a special metadata overlay indicating the EIC Saboteurs	I can use this secret information to guide the Nawab's Loyalists	As a Siraj ud-Daulah, I want to see a special metadata overlay indicating the EIC Saboteurs, so that I can use this secret information to guide the Nawab's Loyalists.	- Spies are highlighted with a red outline on the player list for Siraj. - The specific roles of individual spies (e.g., Mir Jafar) are not shown to balance the game. - The vision overlay is persistent throughout the entire game.	Low	To Do
7	FR_5	The system shall reveal faction-specific metadata to authorized roles.	US_6	EIC Spy Mutual Recognition	EIC Spy	to see my fellow EIC spies highlighted on the game screen	we can coordinate our sabotage strategy without exposing ourselves	As a EIC Spy, I want to see my fellow EIC spies highlighted on the game screen, so that we can coordinate our sabotage strategy without exposing ourselves.	- All EIC spies see red badges next to each other's names on the player grid. - No Loyalist player name cards contain any indicators. - Siraj ud-Daulah's card is not highlighted to the spies.	Medium	To Do
8	FR_6	The system shall moderate a phase-driven discussion loop.	US_8	Discussion Phase Timer	Player	to see a countdown timer during the discussion phase	I know how much time is left to debate and accuse other players	As a Player, I want to see a countdown timer during the discussion phase, so that I know how much time is left to debate and accuse other players.	- A visible radial countdown timer is displayed at the top of the UI. - The timer plays a ticking sound effect during the final 5 seconds. - The game state automatically transitions to the Minister Selection phase when the timer reaches zero.	Medium	In Progress
9	FR_7	The system shall orchestrate a stateful voting engine for proposed teams.	US_9	Cast Ballot Vote	Player	to cast a secret "Yes" or "No" vote on the proposed team	I can express my trust or distrust in the selected ministers	As a Player, I want to cast a secret "Yes" or "No" vote on the proposed team, so that I can express my trust or distrust in the selected ministers.	- Two distinct buttons ("Approve" / "Reject") are displayed. - The choice is hidden from other players until all votes are cast. - The vote must be submitted within a 30-second time limit or default to a "Accept" vote.	High	To Do
10	FR_8	The system shall enable active tactical abilities for specialized character roles.	US_10	Trigger de Bussy Ability	de Bussy	to trigger my active ability during the Minister Selection phase	I can inspect a selected minister's proposed card before it is finalized	As a de Bussy, I want to trigger my active ability during the Minister Selection phase, so that I can inspect a selected minister's proposed card before it is finalized.	- An "Activate Ability" button appears once per match. - Upon activation, de Bussy receives a private card-peek screen showing the minister's card. - The target minister is not notified of the inspection.	Medium	To Do
11	FR_7	The system shall orchestrate a stateful voting engine for proposed teams.	US_11	Tally Ballot Results	Player	to see the public ballot results and how the court voted	I can analyze the court voted for which team to deduce the court's alignments	As a Player, I want to see the public ballot results , so that I can analyze the court voted for which team to deduce the court's alignments.	- A summary card appears listing the court's votes. - The total counts of Approve vs. Reject are displayed prominently. - If approved, the game transitions to the Resolution Phase; if rejected, the round's fail track increases by one.	Medium	To Do
12	FR_9	The system shall resolve the secret mission cards played by ministers.	US_12	Play Mission Card	Minister	to secretly play a "Success" or "Sabotage" card during the mission resolution	I can work towards my faction's victory objective without revealing my identity	As a Minister, I want to secretly play a "Success" or "Sabotage" card during the mission resolution, so that I can work towards my faction's victory objective without revealing my identity.	- Nawab's Loyalists are Prompted into playing the "Success" card. - EIC Spies are given a choice between "Success" and "Sabotage" cards. - The card choice is made on a full-screen overlay with no physical feedback visible to other players.	High	To Do
13	FR_10	The system shall enable lobby moderation capabilities for host accounts.	US_13	Kick Player out from Lobby	Lobby Host	to kick disruptive or inactive players from my lobby	I can maintain a friendly and active group for the match	As a Lobby Host, I want to kick disruptive or inactive players from my lobby, so that I can maintain a friendly and active group for the match.	- A "Kick" button is available next to player names in the lobby setup screen for the host. - Kicked players are disconnected from the room and redirected to the main dashboard. - Kicked players are banned from rejoining that specific lobby.	Low	In Progress
14	FR_11	The system shall support secure user registration and account creation.	US_14	User Account Registration	Unregistered User	to register a new account with a unique username, email, and password	I can create an identity, log in, and save my game progress and match statistics	As an Unregistered User, I want to register a new account with a unique username, email, and password, so that I can create an identity, log in, and save my game progress and match statistics.	- The system validates that the username is unique and contains 3-15 alphanumeric characters. - The system validates that the email is in a valid format and is not already registered. - The password must be at least 8 characters long, containing at least one number and one special character.	High	In Progress
15	FR_11	The system shall support secure user login and session management.	US_15	User Secure Login	Registered Player	to log in securely using my credentials and maintain a persistent session	I can access the matchmaking lobby without needing to log in every time I open the app	As a Registered Player, I want to log in securely using my credentials and maintain a persistent session, so that I can access the matchmaking lobby without needing to log in every time I open the app.	- The login screen provides fields for email/username and password with password masking. - The system generates a secure JWT token stored in the client's local storage for session persistence. - A "Log Out" button clears the token and redirects the player to the login screen.	High	In Progress
16	FR_12	The system shall maintain player profiles and track historical match statistics.	US_16	View Profile and Stats	Player	to view my personal profile page, historical match statistics, and win-loss ratios	I can track my performance over time and see my achievements across different factions	As a Player, I want to view my personal profile page, historical match statistics, and win-loss ratios, so that I can track my performance over time and see my achievements across different factions.	- The profile dashboard displays games played, win rate, and faction-specific win ratios (Loyalist vs. Spy). - The profile page displays the player's level and experience points gained from matches. - Database queries are optimized with Prisma caching for fast profile load times (sub-100ms).	Medium	To Do
17	FR_13	The system shall manage leader rotation and minister team nominations.	US_17	Nominate Minister Team	Leader	to select and nominate a specific number of players to act as ministers for the	the entire court can vote on whether to approve or reject the proposed team	As a Leader, I want to select and nominate a specific number of players to act as ministers for the current mission, so that the entire court can vote on whether to approve or reject the proposed team.	- A selection interface allows checking/unchecking player avatars based on the required team size for the round. - A "Submit Nomination" button becomes active only when the selected player count matches the round's requirements. - The selected player IDs are transmitted to the backend to transition the game state to the Ballot Phase.	High	In Progress
18	FR_3	The system shall support real-time public communication during gameplay.	US_18	Discussion Phase Chat	Player	to send and read text messages in the public chat during the Discussion phase	I can debate, accuse, and coordinate before the team is proposed	As a Player, I want to send and read real-time text messages in the public chat during the Discussion phase, so that I can debate, share suspicions, and coordinate with other players before a team is proposed.	- A text chat input is available to all players during the Discussion phase. - Sent messages appear in real time on every player's screen with sender avatar and username. - The chat module receives the phase-start event and duration so messaging aligns with the active Discussion window.	Medium	In Progress
19	FR_7	The system shall orchestrate a stateful voting engine for proposed teams.	US_19	Track Rejection Count	Player	to see the current consecutive team rejection count (fail track) updated on the screen	I can monitor how close the EIC faction is to winning the round due to deadlocks	As a Player, I want to see the current consecutive team rejection count (fail track) updated on the screen, so that I can monitor how close the EIC faction is to winning the round due to deadlocks.	- A visual track with 5 nodes (1 to 5) is displayed on the game UI. - If a ballot vote fails, the tracker advances by one node and the next player is assigned Leader. - If the tracker reaches 5, the round is automatically awarded to the EIC faction and the tracker resets to 0.	High	To Do
20	FR_5	The system shall reveal faction-specific metadata to authorized roles.	US_20	Mir Madan Round 1 Spy Vision	Mir Madan	to see the identities of all Red team members except Rai Durlabh, during Round 1's discussion phase only	I can gather early intelligence to steer the Green team before the vision is permanently lost	As Mir Madan, I want to see the identities of all Red team members except Rai Durlabh during Round 1's discussion phase, so that I can gather early intelligence to steer the Green team's decisions before the vision permanently expires.	- During Round 1's discussion phase only, Mir Madan's view highlights every Red team member except Rai Durlabh. - Rai Durlabh never appears in this vision, and no other Green player receives the same information. - The vision is server-removed the moment Round 1 ends; Rounds 2–5 show Mir Madan no special information.	High	To Do
21	FR_8	The system shall enable active tactical abilities for specialized character roles.	US_21	Jean Law Win-Credit Lock	Jean Law	to lock onto one Red team member at any point during the game	the locked member is denied win credit even if Red wins, letting me shape who benefits from a Red victory	As Jean Law, I want to lock onto one Red team member at any point during the game, so that the locked member receives no win credit even if Red wins the round or the game.	- Jean Law has an active "Lock" action usable once, targeting any single Red team member; once locked, the target cannot be changed (- The locked member is excluded from win credit at endgame even if the Red team wins the round or the match. - The lock is a deliberate active choice, not automatic, and the target is not notified of being locked.	Medium	To Do
22	FR_14	The system shall compile match results and evaluate faction victory conditions.	US_22	Jagat Seth Endgame Guess	Jagat Seth	to make one endgame guess identifying a specific hidden player, based on which side won the majority of rounds	I can win the game outright for myself, overriding the normal Green/Red result	As Jagat Seth, I want to make a single endgame guess — Rai Durlabh if Red won the majority, or Mir Jafar if Green won the majority — so that I can win the game outright for myself and make everyone else lose.	- The guess prompt appears only in the Endgame phase, only when the player count includes a Gold slot, and allows exactly one guess w - If Red won the majority the target is Rai Durlabh; if Green won the majority the target is Mir Jafar, and this branch runs only after Mir Ja - A correct guess makes Jagat Seth the sole winner and marks all other players (Green and Red) as losers; a wrong guess is discarded a	Medium	To Do
23	FR_8	The system shall enable active tactical abilities for specialized character roles.	US_23	Trigger St. Frais Mission Peek	St. Frais	to inspect one of the played mission cards before they are shuffled and revealed	I can gather intelligence on which minister played a "Sabotage" or "Success" car	As a St. Frais, I want to inspect one of the played mission cards per game, so that I can gather intelligence on which minister played a "Sabotage" or "Success" card.	- St. Frais (an EIC Spy) gets a "Peek Played Card" option during the Resolution phase if he is not on the mission. - The peek action displays the card content (Success/Sabotage) of a chosen minister in a private overlay. - The selected minister is not notified of the card-peek event.	Medium	To Do
24	FR_14	The system shall compile match results and evaluate faction victory conditions.	US_24	Compile Scoreboard and Win State	Player	to see the updated match score and victory announcement when a faction reaches 3 wins	I know who won the match and the game session is officially concluded	As a Player, I want to see the updated match score and victory announcement when a faction reaches 3 wins, so that I know who won the match and the game session is officially concluded.	- A scoreboard displays the outcomes of each completed round (Nawab Success or EIC Sabotage). - When either Nawab's Faction or EIC Faction wins 3 rounds, a full-screen victory overlay is triggered. - The victory screen displays a thematic historical summary of the outcome (e.g., Nawab preserves independence or EIC establishes dominance).	High	In Progress
25	FR_15	The system shall display a post-match breakdown and update player profiles.	US_25	View Post-Game Report	Player	to view a comprehensive post-match report detailing all player roles, votes, and mission contributions	I can analyze players' strategies, verify suspicious behaviors, and discuss the match	As a Player, I want to view a comprehensive post-match report detailing all player roles, votes, and mission contributions, so that I can analyze players' strategies, verify suspicious behaviors, and discuss the match.	- The post-game lobby reveals all player names mapped to their true roles (e.g., Mir Jafar, Jagath Seth, Siraj ud-Daulah). - A round-by-round timeline shows who was nominated, how everyone voted on the ballot, and the mission cards played. - A chat box remains active in the post-game lobby for discussion.	Medium	In Progress
26	FR_19	The system shall manage player lobby departures and host migration.	US_26	Lobby Host Migration	Player	to leave a game lobby, and if I am the host, have the lobby ownership migrate to another player	the lobby remains active and other players can still start the game	As a Player, I want to leave a game lobby, and if I am the host, have the lobby ownership migrate to another player, so that the lobby remains active and other players can still start the game.	- If a non-host player leaves the lobby, their slot is freed, and the lobby count updates for all other players. - If the host leaves, the system assigns the oldest connected player in the room as the new host. - The new host is notified of their status and gains access to the "Lobby Settings" and "Start Game" buttons.	Medium	In Progress
27	FR_16	The system shall deliver dynamic, immersive audio-visual content.	US_27	Dynamic Round Music	Player	to hear ambient audio that dynamically changes intensity based on the game state and round number	I feel a deeper sense of tension and immersion during high-stakes game moments	As a Player, I want to hear ambient audio that dynamically changes intensity based on the game state and round-number, so that I feel a deeper sense of tension and immersion during high-stakes game moments.	- The background music loops at low intensity during the lobby setup and early rounds. - The music transitions to a high-tempo, dramatic orchestral track when either faction has 2 wins or in Round 5. - A setting is available to mute or adjust background music and sound effects volumes individually.	Low	To Do
28	FR_12	The system shall support player profile customization and match history tracking.	US_28	Select Historical Avatar	Player	to select and equip a historical avatar from the profile customization page	I can personalize my appearance to other players during the match setup and gameplay	As a Player, I want to select and equip an avatar from the profile customization page, so that I can personalize my appearance to other players during the match setup and gameplay.	- The profile page contains an avatar gallery with illustrations of key figures (e.g., Mir Madan, Clive, Siraj, Mohan Lal). - Selecting an avatar updates the player's active avatar in the database via Prisma. - The selected avatar is rendered in matchmaking lobbies and during the game's round discussion.	Low	In Progress
29	FR_17	The system shall manage player network state and handle reconnections gracefully.	US_29	Player Reconnection Handling	Player	to reconnect to an active match if my connection drops and resume my role without disrupting the game	I can finish the game and avoid losing my progress due to transient network issues	As a Player, I want to reconnect to an active match if my connection drops and resume my role without disrupting the game, so that I can finish the game and avoid losing my progress due to transient network issues.	- Upon disconnection, the system marks the player as \"offline\" on the UI and starts a 60-second grace timer. - If the disconnected player reopens the app, the client retrieves the active game session and restores their state (secret role, cards, etc.). - If the player fails to reconnect within the timer, they are auto-defeated, and the match is terminated or resolved.	High	To Do
30	FR_18	The system shall enforce server-side validation of game rules and state transitions.	US_30	Server-Side Rule Enforcement	Player	every game action (nominations, votes, card plays) to be authoritatively validated by the server	no player can cheat or perform out-of-order moves by tampering with the client-side application	As a Player, I want to have all game actions (nominations, votes, card plays) validated on the server, so that no player can cheat or perform out-of-order moves by tampering with the client-side application.	- The server implements a finite state machine using Redis state caching to track the active phase and expected actions. - Any incoming socket messages (e.g., cast_vote, play_card) that do not match the expected state are rejected with an error. - The server verifies that only the designated Leader can submit nominations and only current Ministers can play cards.	High	To Do